

Aiden J. Eilertson

2846 220th St. N, Hawley, MN 56549 | (218) 443-4670 | aeilert37@gmail.com | LinkedIn Profile:

www.linkedin.com/in/aiden-eilertson-5ba315337/

EDUCATION

Bachelor of Science in Biomedical Engineering

December 2027

University of North Dakota, Grand Forks, ND

Current GPA: 4.0 / 4.0

- Mechanical Engineering Concentration
- Minor in Pre-Medicine

EXPERIENCE

Medical Surgical Nursing Assistant

September 2025 - Present

Altru Hospital, Grand Forks, ND

- Provide compassionate care for patients in the hospital setting
- Monitor, document, and communicate patients' health changes

Mechanical Engineering Intern & Student Assembly Technician

May 2022 – August 2026

PharmaMed Automation, Fargo, ND

- Design/modify parts for custom pharmaceutical machines using CREO
- Create drawings for parts and assemblies
- Assist in machine assembly

Math Tutor

August 2025 – May 2026

University of North Dakota, Grand Forks, ND

- Assist students with math & proctor student exams

NOTABLE PROJECTS / RESEARCH

Non-Contact Tympanic Membrane (TM) Monitoring

August 2026 – Present

University of North Dakota, Grand Forks, ND

- Researching accessible and miniaturized method of monitoring TM vibration without physical contact

Permanent Scleral Lens Marking

January 2026 – May 2026

University of North Dakota, Grand Forks, ND

- Worked to develop a method of permanently marking an RGP scleral lens for orientation

Bone Fracture Prediction

January 2025 – December 2025

University of North Dakota, Grand Forks, ND

- Developed a machine learning-driven finite element modeling framework to simulate bone stress distributions and predict patient-specific fracture risk for improved clinical assessment
- Presented findings via abstract and poster at the 2025 Biomedical Engineering Society Annual Meeting

Wireless ECG Signal Quality Analysis

August 2024 - December 2024

University of North Dakota, Grand Forks, ND

- Built a 3-lead benchtop ECG testing system to evaluate signal transmission quality between electrodes at controlled distance intervals, which advanced research in wireless ECG development
- Obtained IRB approval, presented research at the 62nd Rocky Mountain Bioengineering Symposium, and published findings in Volume 61 of the Biomedical Sciences Instrumentation journal

TECHNICAL SKILLS

Python | MATLAB | CREO | Finite Element Analysis | Machine Learning | Prototyping | Assembly | Instrumentation

HONORS & LEADERSHIP

University of North Dakota

- Vice President, Biomedical Engineering Society Student Chapter
- Member, Tau Beta Pi Engineering Honor Society
- Barry Scholar, The Barry Foundation
- Recipient, Presidential Scholarship
- Leadership Team Member, Cru

Hawley High School

- Recipient, American Legion School Award & Citizenship Award
- Member, Student Council (4 years) & Minnesota Honor Society (3 years)